PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A systematic review on the use of patient-held health records in low and middle-income countries.
AUTHORS	Joseph, Linju; Lavis, Anna; Greenfield, Sheila; Boban, Dona; Humphries, Claire; Jose, Prinu; Jeemon, Panniyammakal; Manaseki-Holland, Semira

VERSION 1 – REVIEW

REVIEWER	Garfield, Sara recommended Imperial College Healthcare NHS Trust
REVIEW RETURNED	09-Dec-2020

GENERAL COMMENTS	This is well written review overall and addresses a clear research gap with practical implications. I have some minor suggestions: 1. There are some grammatical errors and typos such as (n=x). Please proof read carefully. 2. Define any abbreviations used in the abstract. I could not understand the abstract fully until having some of the abbreviations defined in the full text. 3. Be clear and consistent throughout about whether you are including communicable diseases. 4. Please add details of who did the initial data extraction (before the tables were reviewed). 5. Try and synthesize the findings from the different studies more in each subsection of the results.

REVIEWER	Jaakkimainen, R. Liisa Institute for Clinical Evaluative Sciences
REVIEW RETURNED	13-Jan-2021

GENERAL COMMENTS	This is an important systematic review on the use of personal health records (paper based) in low- and middle-income countries on patient and provider satisfaction and healthcare use. My comments are minor. Abstract Please spell out the abbreviations (PHCs and LMICs) Introductions The introduction is well written and clear. Methods The methods are well written and include a quality assessment of the articles. 1. Was a librarian used to help identify article searches? 2. While I appreciate it's difficult to know what outcomes (satisfaction or healthcare use) were the focus of the studies before starting the systematic review, maybe clearly articulating what would be important for the systematic review in the methods section would

	strengthen conclusions (see my comment below)? Table 1 is good, but I don't see intermediate healthcare outcomes such as medication reconciliation or the results of lab test on the list. 3. The reference section of review articles were examined for further studies. But were the references of the selected studies also reviewed to identify more studies? Results 4. I think the numbers are to be provided for the number or studies included? "There were studies from Zambia (n=x), Lesotho (n=x), South Africa (n=x), China (n=x), and Mongolia (n=x)…" 5. Maybe add the 3 studies found in the grey literature into the results section versus discussion section. 6. The study selection figure is good. Maybe more information/details on why the vast number of articles were excluded? Discussion/Conclusions I think this study did identify some important next steps for future work. This includes publishing more high-quality studies (or at least including information from their methods in order that study quality can be assessed). I would suggest being a little more directive in what should be the content of PHRs or more specific healthcare outcomes so that PHRs could ultimately improve the transfer of relevant information between providers or health sectors that lead to quality care. The authors mention a body of research on PHR for maternal/childcare in low- and middle-income countries. What components of the content of PHR care in this instance improve care? I understand that for PHRs to be used and accepted, a demonstration of patient and provider satisfaction is needed. But I would like to see the conclusions to guide new research studies to move past satisfaction outcomes and identify specific components in PHRs (medications, lab results, BPs) for chronic disease management and also specific healthcare outcomes.

REVIEWER	Al-Sallakh, Mohammad Swansea University Medical School
REVIEW RETURNED	25-Jan-2021

GENERAL COMMENTS	This systematic review summarises the evidence about the use and utility of non-pediatric, non-maternity, patient-held health records (PHRs) in low- and middle-income countries (LMICs) and the views of health care professionals and patients about PHR use. This is a vital topic about barriers of continuity of care in LMICs. However, the included studies did not examine the impact of PHRs on health outcomes (whether objective or patient-reported), which is an important topic of further research, as is the safety and validity of the information recorded in PHRs. I have minor comments: - In the abstract, I suggest the abbreviations 'PHR' and LMICs to be defined at the first use. - In the 'Results/Characteristics of included studies', I am not sure what 'n=x' means next to the country name. - In 'Synthesis of results', the abbreviation PHB was not in the manuscript. Does it refer to 'patient health booklet' or similar? Or is it a typo for PHR?

	- Few typos: e.g., Google scholar –> Google Scholar

VERSION 1 – AUTHOR RESPONSE

Comments from Reviewer: 1
This is well written review overall and addresses a clear research gap with practical implications. I have some minor suggestions:
Response
We appreciate the positive feedback from the reviewer.

1. There are some grammatical errors and typos such as (n=x). Please proof read carefully.
Response
Thank you for pointing out the typos and grammatical errors. We have made the necessary changes throughout the manuscript.

2. Define any abbreviations used in the abstract. I could not understand the abstract fully until having some of the abbreviations defined in the full text.
Response
Thank you for pointing this out. We have expanded the abbreviations used in the abstract page number 3, line numbers 36-39.
"To review the available evidence on the benefit of patient-held records (PHRs), other than maternal and child health records for improving the availability of medical information for handover communication between healthcare providers (HCPs) and/or between HCPs and patients in low and middle income countries (LMIC)."

3. Be clear and consistent throughout about whether you are including communicable diseases.
Response
We have excluded the studies describing patient-held records for single communicable disease such as tuberculosis. The following has been included on page 7, Table 1 Inclusion and exclusion criteria.

4. Please add details of who did the initial data extraction (before the tables were reviewed).
Response
We have added the following sentence on page number 9, data extraction section, line numbers 193-194.
"Two reviewers (DB and PJ) independently extracted data from the included studies. LJ reviewed all the data extraction tables."

5. Try and synthesize the findings from the different studies more in each subsection of the results.
Response
We thank the reviewer for this useful suggestion. We have made the following changes:
On page number 17, section patients carrying PHRs , line number 283 "Overall, most patients carried their PHRs to healthcare visits."; page number 17 line number 299, "The HCPs recording in PHRs was sub-optimal."; page number 18, line number 308, "Generally, HCPs perceived PHRs to be useful in recording patients' medical information." In addition, on page number 18, line number 326 "The studies reported very limited protocols and training available for HCPs on the use of PHRs." Finally on page number 19, line number 337 "…and in general, found that patients regarded PHRs as important documents containing their own medical information."

Comments from Reviewer: 2

This is an important systematic review on the use of personal health records (paper based) in low- and middle-income countries on patient and provider satisfaction and healthcare use. My comments are minor.
Response
We appreciate the reviewers' accurate summary of the paper.

1. Abstract
Please spell out the abbreviations (PHCs and LMICs)
Response
Thank you for pointing this out. Kindly refer to the changes made in the section above.

2. Introductions
The introduction is well written and clear.
Response
We appreciate the positive feedback from the reviewer.

3. Methods
The methods are well written and include a quality assessment of the articles.
1. Was a librarian used to help identify article searches?
Response
We did not use a librarian to identify article searches. However, authors experienced in handover communication (SMH) and systematic review (JP) reviewed the search terms.

2. While I appreciate it's difficult to know what outcomes (satisfaction or healthcare use) were the focus of the studies before starting the systematic review, maybe clearly articulating what would be important for the systematic review in the methods section would strengthen conclusions (see my comment below)? Table 1 is good, but I don't see intermediate healthcare outcomes such as medication reconciliation or the results of lab test on the list.

Response
Thank you for this excellent suggestion. We have incorporated the following in the table 1, page 8 in the outcomes section.
"Intermediate outcomes such as blood pressure/blood glucose monitoring, medication management/reconciliation and/monitoring of lab values were included."

3. The reference section of review articles were examined for further studies. But were the references of the selected studies also reviewed to identify more studies?
Response
We have examined the references of selected studies. No new studies were included after the review of references of the selected studies.

4. Results
I think the numbers are to be provided for the number or studies included? "There were studies from Zambia (n=x), Lesotho (n=x), South Africa (n=x), China (n=x), and Mongolia (n=x)…"
Response
Thank you for pointing this out. We have made the changes in the manuscript; please refer to the response to reviewer 1 above.

5. Maybe add the 3 studies found in the grey literature into the results section versus discussion section.
Response
Thank you for this suggestion. However, we believe that since the studies found through grey

literature (n=3) do not include description or evaluation of the patient-held record, it would not contribute to the aim of understanding the use of patient-held records for the review. Hence, we feel it is more appropriate to include these in the discussion section. To do otherwise would draw attention to the lack of studies describing/evaluating the patient-held records, which are routinely being used in low and middle-income countries.

6. The study selection figure is good. Maybe more information/details on why the vast number of articles were excluded?
Response
We have added the following details on the excluded studies on page 12, in the results section, line numbers 288-233 :
“We excluded 115 articles including: studies that described PHRs but were not conducted in LMICs (56); studies that described maternal or child health records in LMICs (29); patient-health education booklet for NCDs (3); studies that described electronic health record implementation (21) and studies that described facility based records (12).”

7. Discussion/Conclusions
I think this study did identify some important next steps for future work. This includes publishing more high-quality studies (or at least including information from their methods in order that study quality can be assessed). I would suggest being a little more directive in what should be the content of PHRs or more specific healthcare outcomes so that PHRs could ultimately improve the transfer of relevant information between providers or health sectors that lead to quality care. The authors mention a body of research on PHR for maternal/childcare in low- and middle-income countries. What components of the content of PHR care in this instance improve care? I understand that for PHRs to be used and accepted, a demonstration of patient and provider satisfaction is needed. But I would like to see the conclusions to guide new research studies to move past satisfaction outcomes and identify specific components in PHRs (medications, lab results, BPs) for chronic disease management and also specific healthcare outcomes.

Response
We thank the reviewer for this suggestion. We have incorporated the following details in the discussion section-page number 21, paragraph 1, line numbers 390-396:
“Previous HIC studies have shown that use of PHRs for patients with diabetes and hypertension have led to decreases in HbA1c ($p<0.001$)52-54, decrease in diastolic blood pressure ($p<0.05$) 52, improvement in receiving foot examinations (OR: 1.68; 95% CI 1.12 to 2.50)54, having physical activity or exercise advised (OR: 1.84; 95% CI 1.16 to 2.92) 54, monitoring of weight, blood glucose and cholesterol52. Similar results may be achieved in LMICs for monitoring blood pressure/glucose/cholesterol, improving foot/eye examination if PHRs are well documented.
And on page 22, paragraph 2, line numbers 427-429.
“Therefore, PHRs having a minimum of details such as diagnosis, medication, lab results and follow-up will enable better information transfer in the health systems transitioning to electronic health records.”

Comments from Reviewer: 3

Comments to the Author:
This systematic review summarises the evidence about the use and utility of non-pediatric, non-maternity, patient-held health records (PHRs) in low- and middle-income countries (LMICs) and the views of health care professionals and patients about PHR use.

This is a vital topic about barriers of continuity of care in LMICs. However, the included studies did not examine the impact of PHRs on health outcomes (whether objective or patient-reported), which is an important topic of further research, as is the safety and validity of the information recorded in PHRs.
Response
We thank the reviewer for accurately summarising and highlighting the important findings from the review.

I have minor comments:

1- In the abstract, I suggest the abbreviations 'PHR' and LMICs to be defined at the first use.
Response
Thank you for pointing this out. Kindly refer to the changes outlined in the response to reviewer 1.

2- In the 'Results/Characteristics of included studies', I am not sure what 'n=x' means next to the country name.
Response
Thank you for pointing this out. Kindly refer the changes made in the response to reviewer 1.

3-In 'Synthesis of results', the abbreviation PHB was not in the manuscript. Does it refer to 'patient health booklet' or similar? Or is it a typo for PHR?
Response
Thank you for pointing this out. We have rectified the typo to PHR.

4- Few typos: e.g., Google scholar –> Google Scholar
Response
We have corrected the typos in the manuscript.